

NEUROMUSCULAR TRANSMISSION
IN A MAMMALIAN PREPARATION IN THE ABSENCE OF
BLOCKING DRUGS AND THE EFFECT
OF D-TUBOCURARINE

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SUMMARY

1. Neuromuscular transmission in the presence and absence of D-tubocurarine was examined in cut muscle preparations of the rat diaphragm using intracellular recording techniques. The method of localizing the end-plates is described and evidence is presented indicating that end-plate potential (e.p.p.) recordings were made close to the end-plate region and that reliable estimates of the quantum content of e.p.p.s could be made.

2. During repetitive nerve stimulation at rates between 2 and 10 Hz, e.p.p. amplitudes declined in response to the first 6–8 stimuli to reach a plateau level which was maintained for 1500 stimuli. At higher rates of stimulation some 500–700 stimuli were needed before a plateau level of amplitude was reached. It was demonstrated that these amplitude changes reflected similar changes in the quantal release of ACh.

3. Tubocurarine in a concentration of 4×10^{-8} g/ml. had no effect on quantal release but did reduce quantal size. In the presence of tubocurarine, 8×10^{-8} g/ml., small reductions in the quantum content of the first e.p.p. in response to a train of stimuli and the maintained quantal release were observed. In the presence of tubocurarine 4×10^{-7} g/ml., the quantum content of the first e.p.p. and the maintained release were further reduced, to the level previously described for curarized preparations. At this concentration a significant effect of tubocurarine upon the pre-synaptic measurements could be detected (*t* test). Studies on single junctions revealed that the post- and presynaptic effects of tubocurarine developed with different time courses.

4. Neuromuscular preparations *in vivo* have been reported to be much more resistant to maintained stimulation than curarized neuromuscular

preparations *in vitro*. It is suggested that this disparity is explained by the presynaptic action of tubocurarine.

5. From the quantal release rates of the present investigation and reports of ACh release/stimulus/end-plate in previous investigations it can be calculated that a quantum of ACh contains between 12,000 and 21,000 molecules.

INTRODUCTION

Motor nerves are capable of maintained discharges at rates between 5 and 25 Hz and *in vivo* muscle preparations can contract forcibly throughout a prolonged stimulation period (Adrian & Bronk, 1928, 1929; Emmelin & MacIntosh, 1956). Numerous investigators using *in vitro* preparations, however, report that transmitter release declines fairly rapidly at frequencies between 5 and 25 Hz and transmission may fail entirely after several minutes (Dale, Feldberg & Vogt, 1936; del Castillo & Katz, 1954; Elmqvist & Quastel, 1965*a*; Krnjević & Mitchell, 1961; Brooks & Thies, 1962; Robbins & Fischbach, 1971).

This discrepancy might be explained if the fatigue of neuromuscular transmission which occurs in isolated preparations was due to some special feature of the experimental situation. Quantal release estimates from preparations stimulated for prolonged periods for example have been made from preparations exposed to tubocurarine, a neuromuscular blocking agent (Krnjević & Mitchell, 1961; Brooks & Thies, 1962; Robbins & Fischbach, 1971).

Tubocurarine attaches to acetylcholine (ACh) receptors on the post-junctional membrane and reduces the amplitude of the e.p.p. (Langley, 1905; Dale *et al.* 1936; Fatt & Katz, 1951). A controversy arises over whether this is the only site of tubocurarine action. This widely used neuromuscular blocking agent was originally thought to have a pre-synaptic effect (Bernard, 1856) and the possibility has been reconsidered by a number of investigators. The prevailing view is that tubocurarine has no effect on transmitter release from nerve terminals (Dale *et al.* 1936; Krnjević & Mitchell, 1961; Beranek & Vyskocil, 1967; Bowen & Merry, 1969; Fletcher & Forrester, 1970; Auerbach & Betz, 1971; Bauer, 1971) but there have been persistent reports that tubocurarine does reduce the amount of transmitter released by nerve impulses (Lilleheil & Naess, 1961; Beani, Bianchi & Ledda, 1964; Standaert, 1964; Riker & Okamoto, 1969; Blaber, 1970; Galindo, 1971).

The rat diaphragm cut muscle preparation (Barstad, 1962) offers a means of examining the quantal release of ACh in the absence of blocking agents. Using this preparation we have examined the effect of short periods (40 impulses) of nerve stimulation upon e.p.p. amplitudes and

quantal contents in the absence and presence of tubocurarine and of long periods of stimulation (1500 impulses) in the absence of tubocurarine. Brief accounts have appeared (Hubbard, Wilson & Miyamoto, 1969; Wilson & Hubbard, 1970). Recently Auerbach & Betz (1971) criticized our preliminary report (Hubbard *et al.* 1969) on the basis that they could not adequately locate the end-plate regions in cut muscle. They inferred that our results must have been obtained from regions remote from end-plates and that our quantum content estimates were therefore not accurate representations of ACh release. Evidence is presented which demonstrates that these criticisms are not applicable to the present results.

METHODS

Experiments were performed on the left hemidiaphragm and attached phrenic nerve excised from 200 g albino rats. Neuromuscular transmission was blocked by cutting muscle fibres, basically as described by Barstad (1962) except that we cut only the ends of the muscle fibres rather than the more drastic procedure of removing 4–5 mm of muscle tissue on either side of the main intramuscular nerve branch. Our procedure normally left a preparation some 8 mm wide. The experiments were usually commenced 40–50 min after cutting. In agreement with Schuster & Kirsch (1968) the muscle resting potential appears to decline exponentially after cutting. Forty minutes after cutting, the superficial muscle fibres had a resting potential of -32 ± 3.6 mV (mean $\pm$ s.e., $n = 7$) and in these seven experiments the resting potential was monitored for 3.5–4 hr when the mean had fallen to -25 ± 3.6 mV. The preparation was kept in an 8 ml. bath continuously perfused at a rate of 2–3 ml./min with an oxygenated (5% CO₂, 95% O₂) solution containing (mM): NaCl, 137; KCl, 2.5; MgCl₂, 1; CaCl₂, 2; NaH₂PO₄, 1; NaHCO₃, 12; and dextrose, 11, with the aid of a bilateral roller pump. The temperature was maintained at 31–33° C. The reduced potassium content of the bathing fluid was chosen since 5.0 mM-KCl tends to induce nerve block in cut muscle preparations (Randic & Straughan, 1964). Tubocurarine was obtained from the Sigma Chemical Co.

The nerve terminal branches were located with the aid of a dissecting microscope when the diaphragm was transilluminated. The e.p.p.s from a cut muscle fibre characteristically showed a shorter rise time and higher amplitude as the electrode was inserted nearer to the branches as in frog muscle (Fatt & Katz, 1951). We used this technique to localize the end-plate in cut muscle and curarized fibres. Our criterion for beginning an experiment was an e.p.p. rise time of 1.1 msec or less. This often meant that before the experiment began the micro-electrode was inserted several times in a given area to insure its placement in the optimal position. This procedure was greatly facilitated by a Canberra type MC-7 micromanipulator which allowed the electrode tip to be inserted at right angles to muscle fibre surfaces. In some experiments an electrode was inserted into the same cell at different distances from the end-plate. To insure that the same cell was relocated after moving the electrode 50 μ m along the fibre, the resting potential of the fibre was noted and, if needed, the electrode was moved laterally up to 30 μ m. This technique is based on the fact that neighbouring muscle cells have distinctly different resting potentials in cut diaphragm preparations.

The evoked e.p.p.s were recorded intracellularly with glass micro-electrodes filled with 3.0 M-KCl (resistances: 5–15 M Ω). The e.p.p.s were amplified by a Medistor electrometer, displayed on a CRO (Tektronix 565) and photographed on moving film

with a Grass C4 camera. The resting potential of the cell was simultaneously monitored on a d.c. strip chart recorder. The base line noise on the CRO was normally less than $100\ \mu\text{V}$. The difficulty in accurately measuring the amplitude of the e.p.p.s was minimized by using a photographic enlarger and a recording film speed of 250–500 mm/sec. The amplitudes of the e.p.p.s were measured to the nearest 0.05 mV.

Estimate of quantum content

E.p.p.s were evoked by nerve stimulation at various rates between 2 and 150 Hz. There were two types of experiments. When the form of the e.p.p. response and the quantum content was being investigated stimulation was of the appropriate duration to elicit 39–41 e.p.p.s. The 20–39th e.p.p.s were used for calculations of the coefficient of variation (CV). Five serial trains of stimuli at the same frequency with a minute between trains for recovery were always used. Five estimates based on a total of 100 e.p.p.s were thus available for the calculation. When the effect of prolonged stimulation was investigated 1500 stimuli were given to the phrenic nerve. The amplitudes of five groups of 20 e.p.p.s each, from at least 100 consecutive e.p.p.s beginning with the 20th e.p.p. were then used in calculation of the CV.

An equilibrium potential of $-15\ \text{mV}$ was used for the uncut curarized preparation. In cut muscle preparations of the cat an equilibrium potential of $-15\ \text{mV}$ has been reported (Blaber, 1970). In the rat diaphragm cut muscle preparation the low resting potentials suggest an alteration in the concentration of sodium and potassium in the muscle fibres. This change in ionic concentration should alter the equilibrium potential (Takeuchi, 1963). In the rat diaphragm an occasional cut muscle fibre with a low resting potential ($-17\ \text{mV}$) was encountered with a positive e.p.p. which suggests that the equilibrium potential is less than $-15\ \text{mV}$. In addition when the resting potential of a muscle fibre is low ($< -18\ \text{mV}$) the use of an equilibrium potential of $-15\ \text{mV}$ can lead to an over-correction for non-linear summation of the e.p.p.s (Martin, 1956). To avoid this problem an equilibrium potential of $-12\ \text{mV}$ was chosen as the best estimate for the cut muscle preparation. A 3 mV reduction in the equilibrium potential had little effect on the quantal release values. For example the quantal release estimate from a fibre with a resting potential of $-29\ \text{mV}$ was 185 quanta ($\text{CV}^2 = 0.00541$) using an equilibrium potential of $-15\ \text{mV}$ and 193 quanta ($\text{CV}^2 = 0.00518$) using an equilibrium potential of $-12\ \text{mV}$. The mean e.p.p. amplitude from this fibre was 4.53 mV. Correction for non-linear summation (Martin, 1955) resulted in increasing this value to 8.78 mV, using an equilibrium potential of $-15\ \text{mV}$ and 7.48 mV, using an equilibrium potential of $-12\ \text{mV}$.

The e.p.p.s, together with the resting potential of the muscle fibre at the time of recording, were analysed with the aid of a computer programme which applied Martin's (1955) correction and then computed the CV of the 20 e.p.p.s. The quantum size (q) was estimated by multiplying the mean corrected e.p.p. by CV^2 . The five serial estimates of q were pooled and used to calculate the quantal content of the first e.p.p. (m) in each train and the mean quantum content of the last 20 e.p.p.s, referred to as the 'tail' (t), or of the 100 e.p.p.s. An estimation of the mobilization rate can be obtained by multiplying t by the frequency of stimulation. From the quantal content of the first 5–6 e.p.p.s the available store of quanta (n) was calculated according to Elmqvist & Quastel (1965*a*) with the precautions previously described (Hubbard, Jones & Landau, 1971) and this quantity divided into m yielded the fraction of the store released by the first impulse which is a measure of the probability of release (p). The ratio of the amplitude of the first e.p.p. relative to the tail (t/m) was calculated as a quantitative measure of the depression of quantal release relative to the first e.p.p.

RESULTS

End-plate potentials and miniature end-plate potentials (m.e.p.p.s)

During the first 20–30 min after cutting as previously reported (Lilleheil, 1965) the resting potentials are usually high enough (-35 to -50 mV) to record m.e.p.p.s and in five experiments e.p.p.s and m.e.p.p.s were recorded at the same junction. The mean m.e.p.p. frequency in these five experiments was 3.4/sec. The rise time of e.p.p.s recorded from cut muscle

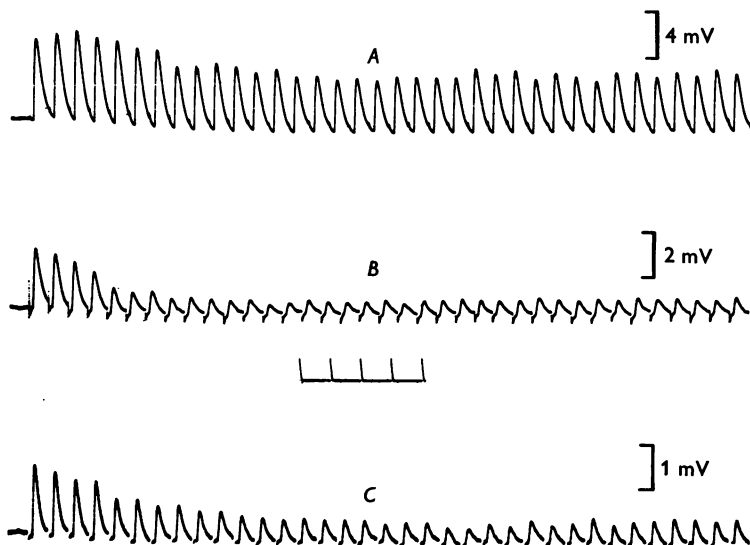


Fig. 1. The effect of D-tubocurarine on the pattern of e.p.p.s during repetitive stimulation at 150 Hz. *A*, e.p.p.s from a cut muscle fibre in the absence of tubocurarine. *B*, e.p.p.s from an uncut muscle fibre exposed to tubocurarine 1.1×10^{-6} g/ml. *C*, e.p.p.s from a cut muscle fibre exposed to tubocurarine 4×10^{-7} g/ml. The time marker in *B* indicates 10 msec intervals.

was similar to the rise time of e.p.p.s recorded from uncut curarized preparations. In five experiments in which a micro-electrode was inserted at different points along the same muscle fibre the minimum rise time was either 0.9 msec (three experiments) or 1.0 msec (two experiments). The amplitude of the e.p.p. was positively correlated with the resting potential of the fibre and usually ranged between -3 and -10 mV. A slight hyperpolarization was characteristically observed on repetitive nerve stimulation in the cut muscle (0.5–1.0 mV) and curarized uncut (0.2–0.5 mV) muscle preparations. The explanation for this is unknown.

As previously reported (Lilleheil, 1965) the pattern of e.p.p.s recorded in response to repetitive stimulation (Fig. 1 *A*) was different from that seen

in diaphragms paralysed by tubocurarine (Fig. 1*B*). The first 2 or 3 e.p.p.s during stimulation at 50–150 Hz were often of the same size and although later e.p.p. responses were smaller, the plateau size obtained was larger than in curarized preparations stimulated at this frequency.

Prolonged stimulation

During prolonged stimulation at frequencies between 2 and 20 Hz there was often no significant change in e.p.p. amplitudes when the initial stimulation period (20–119th e.p.p.) was compared with the end of the stimulation period (1400–1499th). For example, as Fig. 2 shows, the mean

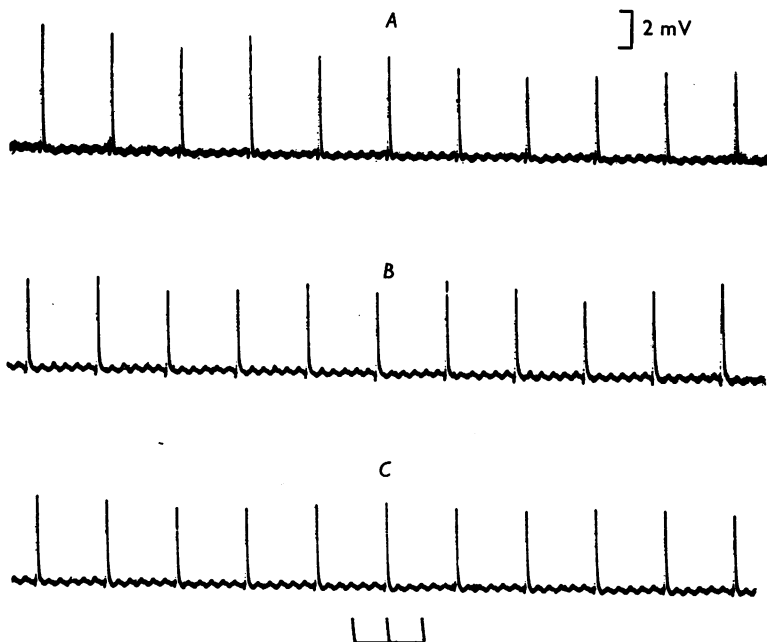


Fig. 2. The effect of prolonged stimulation on e.p.p. amplitude in a cut muscle preparation. Phrenic nerve stimulation at 10 Hz was continued for a total of 1500 stimuli. *A* shows the first 11 e.p.p.s in the train. *B* shows the 700–710th e.p.p. and *C* shows the 1400–1410th e.p.p. in the train. All recordings are from the same junction and trial. Calibration markers indicate 50 msec and 2 mV.

amplitude of the 7–11th e.p.p. (Fig. 2*A*) at 10/sec stimulation was essentially the same as the mean amplitude of the 700–710th e.p.p.s (Fig. 2*B*) and the 1400–1410th e.p.p.s (Fig. 2*C*). The only significant decline in e.p.p. amplitude noted at this junction occurred during the first 6 evoked e.p.p.s where the amplitude declines 34 % compared with the first e.p.p. (Fig. 2*A*). During prolonged stimulation at higher frequencies (50–150 Hz)

at this and other junctions, the amplitude of the e.p.p.s declined 38–48 % in the later stimulation periods when compared with the initial stimulation period (e.p.p.s 20–119).

Measurements of quantum content

The results illustrated in Figs. 1 and 2 suggested there was a difference between the properties of e.p.p.s in cut muscle preparations and the properties of e.p.p.s in curarized preparations, which could be explained if there was a fall in quantum content in curarized preparations.

TABLE 1. Influence of distance from end-plate on e.p.p. parameters in cut muscle. The point from which the e.p.p. with fastest rise time and largest amplitude was recorded has been arbitrarily assigned 0 position. Other distances are measured to either side ($\pm$) of this spot. The mean quantum content of the first e.p.p. (m), the 20–39th e.p.p.s (t), and the releasable store (n) are given in numbers of quanta. The quantum size (q) is expressed in mV

Distance (μm)	From single stimuli		From train of stimuli			
	Rise time (msec)	Amplitude (mV)				
			$q(\text{mV})$	m	t	n
– 50	1.0	1.69	0.019	104	46	3554
0	0.9	2.61	0.025	127	51	4159
+ 50	1.0	2.23	0.024	110	46	3405
+ 100	1.1	1.86	0.017	133	54	2678
+ 150	1.2	1.03	0.012	118	47	2448

There are, however, certain obstacles to quantum content measurements in cut muscle preparations which have been well outlined by Auerbach & Betz (1971). These workers suggest that when muscle fibres of rat diaphragms are cut to block twitching there is a reduction in the muscle fibre space constant to the order of 400–500 μm , so that junctional signals are distorted by the cable properties of the muscle fibres. It is known that the space constant of the uncut rat diaphragm muscle is between 380 and 640 μm (Zolovick, Norman & Fedde, 1970). It seems doubtful therefore whether the space constants of the rat diaphragm muscle fibres are reduced when the muscle fibres are cut. The question as to distortion of e.p.p.s due to a small space constant however remains. Whether recordings are distorted or not presumably depends on the closeness of the recording site to the end-plate. We adopted as our criteria of closeness a measurement of e.p.p. rise time of 1.1 msec. This decision was based on experiments such as those illustrated in Table 1 in which an electrode was inserted into the same fibre at different points along its length and thus at differing distances from the end-plate. In the five experiments of this type

the shortest rise time and greatest amplitude of e.p.p.s occurred together as expected (Fatt & Katz, 1951; Nickels, 1970). In these experiments it was also possible to carry out quantum content measurements with results such as those illustrated in Table 1 (train of stimuli). It will be noted that a comparison of columns 2 and 3 in Table 1 indicates that the end-plate lay between positions 0 and +50. Estimates of q were largest at these two

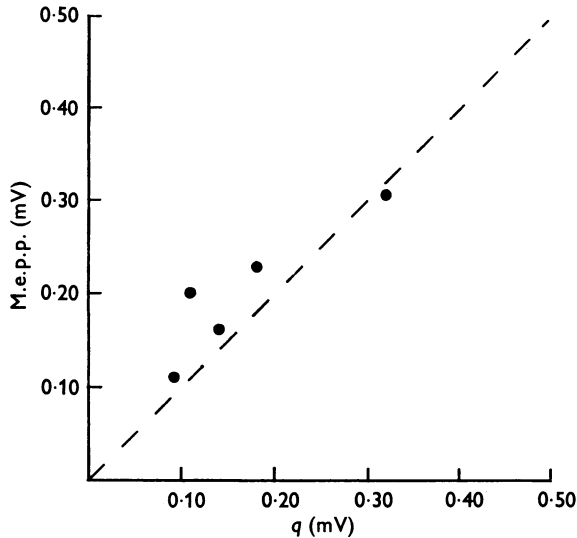


Fig. 3. Comparison between m.e.p.p. amplitude and quantum size (q) estimated from five different junctions. The quantum size, q , was calculated from the coefficient of variation of five groups of 20 e.p.p.s in the tail of a train (20–39th e.p.p.s). The dashed line corresponds to equality for these two estimates.

points and progressively smaller at more distant points. The quantum estimates of m and t at several recording distances from the end-plate was of particular interest since Auerbach & Betz (1971) hypothesized that these values should be over-estimated when the recording electrode is located 100 μm or farther from the end-plate focus. Since such distortions were not observed at recording sites up to 150 μm from the end-plate focus the seriousness of exact electrode placement appears to be less than predicted. From the data given in Table 1 and from similar experiments we were able to calculate a space constant for the cut muscle fibres, although the inherent variability of e.p.p. amplitudes of course limits the accuracy of the estimates, which ranged between 300 and 450 μm , in essential agreement with Zolovick *et al.* 1970.

In subsequent experiments the utility of our criteria was further tested

in two situations where the effect of space constant on e.p.p. amplitudes would be expected to affect our results.

First, we carried out five experiments within 25–30 min of cutting the muscle fibres. In these experiments we measured m.e.p.p. amplitudes and quantum size from trains of e.p.p.s at 100 Hz. As Fig. 3 shows there was a satisfactory agreement in three of the five experiments between the two estimates of quantum size. In the other two experiments, either the estimate of q from variance measurements was too small or the mean m.e.p.p. amplitudes were too large. We are inclined to the latter view as it was difficult to distinguish small m.e.p.p.s from the base line noise in these experiments due to the low resting potential of the muscle fibres.

TABLE 2. Effect of Ca^{2+} on the mean end-plate potential amplitude in the tail of the train (e.p.p. 20–39) and estimation of quantum size (q) in cut muscle. The e.p.p.s in the tail have been corrected for non-linear summation and were produced by trains of stimuli at 150/sec for 0.3 sec. All figures are in mV. Each end-plate was subjected to a low (1.5 mM) and high (3.0 mM) Ca solution

End-plate	3.0 mM Ca^{2+}		1.5 mM Ca^{2+}		
	q_1	E.p.p. 20–39	q_2	E.p.p. 20–39	q_1/q_2
AA	0.021	5.00	0.018	2.73	1.17
BB	0.007	2.33	0.008	1.44	0.88
CC	0.012	3.86	0.013	2.75	0.92
DD	0.007	3.10	0.008	2.36	0.88
EE	0.023	5.46	0.020	3.09	1.15
FF	0.021	4.75	0.022	3.45	0.95
Mean $\pm$ s.e.					0.99 $\pm$ 0.02

Secondly, in six experiments we deliberately altered e.p.p. amplitudes at a given junction during an experiment by changing the extracellular $[\text{Ca}^{2+}]$ from 1.5 to 3 mM or carrying out the reverse procedure. We compared our estimates of quantal size in the two situations when the e.p.p. amplitudes had stabilized in the new $[\text{Ca}^{2+}]$. As Table 2 shows, the change in e.p.p. amplitudes varied from 31 to 83 % but there was no significant change in the estimates of q . We concluded that we could safely proceed to measure quantum content and other related parameters.

E.p.p. parameters in cut-muscle

The mean quantum content of the first e.p.p. in Table 3 (237 ± 19) is similar to that reported by workers using curarized diaphragm muscles (e.g. Hubbard *et al.* 1971). The mean value for n ($5,549 \pm 547$) is however much larger than previously reported (e.g. 375, Elmqvist & Quastel, 1965*a*), and the p values (0.049 ± 0.004) are much smaller than the 0.13

reported by Elmqvist & Quastel (1965*a*). Furthermore, mobilization, which can be obtained by multiplying t in Table 3 by 150 (21,600) is much higher than the 2500 quanta/sec reported by Elmqvist & Quastel (1965*a*). Furthermore, during prolonged stimulation we found that quantum content was much better maintained than had been previously reported (Elmqvist & Quastel, 1965*a*; Robbins & Fischbach, 1971). At all frequencies there was a fall in quantum content of the e.p.p.s (20–39) when

TABLE 3. Effect of tubocurarine on e.p.p. parameters (mean $\pm$ 1 s.e.) in cut muscle. The e.p.p.s were evoked by trains of stimuli at a frequency of 150 Hz. The values for the first e.p.p. (m), the mean output by the 20–39th e.p.p.s (t) and the releasable store (n) are expressed in quanta. The quantum size (q) is expressed in mV. The probability of release (p) and the ratio of the 20–39th e.p.p.s relative to the first (t/m) are dimensionless fractions. N , indicates the number of end-plates sampled in the study. * indicates significant difference ($P < 0.01$ in t test) between control and treated values

$N =$	Control	Tubocurarine	
		8×10^{-8} g/ml.	4×10^{-7} g/ml.
	18	13	18
q	0.030 ± 0.005	0.019 ± 0.004	$0.009 \pm 0.001^*$
m	237 ± 19	212 ± 14	$165 \pm 17^*$
t	144 ± 12	115 ± 8	$55 \pm 3^*$
t/m	0.61 ± 0.02	0.55 ± 0.03	$0.37 \pm 0.03^*$
n	5549 ± 547	4965 ± 469	$1690 \pm 164^*$
p	0.049 ± 0.004	0.048 ± 0.006	$0.112 \pm 0.016^*$

compared to the first e.p.p. which ranged from 20% at 2 Hz to 40–50% at frequencies of 100–150 Hz. Thereafter two patterns of response were found. As Fig. 4 shows, at frequencies of 10 or less there was no significant further decline in quantum content when e.p.p.s 20–119 were compared with e.p.p.s 1400–1499. At higher frequencies there was a decline, the magnitude of which varied greatly at individual junctions. For instance, we found two junctions stimulated at 20 Hz and one junction stimulated at 50 Hz which showed no decline in e.p.p. quantum contents. Fig. 4 does not make clear whether the decline of quantum content normally found during stimulation at frequencies above 10–20 Hz was continuous or occurred early or late in the tetanic train. Fig. 5 illustrates the whole time course of the changes in quantum content at one junction stimulated at 150 Hz. The decline of quantum content occurred as the result of the first 400–500 stimuli and thereafter quantal contents were maintained. Eight other experiments at frequencies of 50 Hz (four), 100 Hz (two), and 150 Hz (two) showed the same result. As a further check, experiments were carried out with a range of frequencies at the same junction. As Table 4 shows, in all experiments of this type the pattern of response were similar

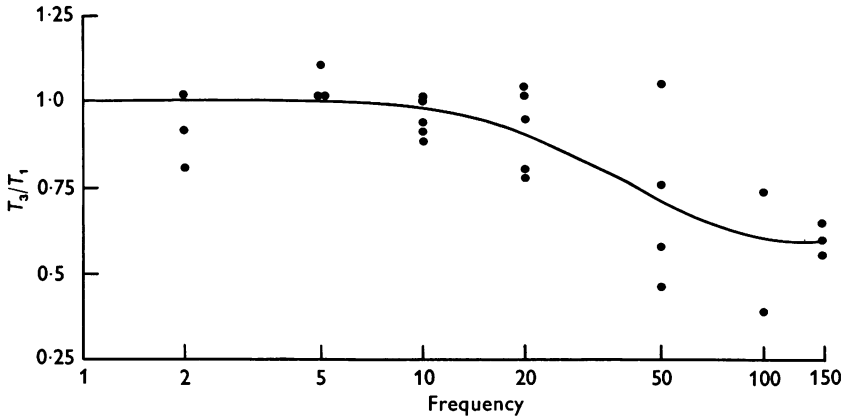


Fig. 4. The effect of frequency of stimulation on the degree of neuromuscular depression during prolonged nerve stimulation. Abscissa: frequency of stimulation (Hz). Note log scale. Ordinate: the depression of quantal output expressed as the ratio of the mean quantum release during the end of the stimulation period, T_3 (1400–1499th e.p.p.) with the mean quantum release during the initial stimulation period, T_1 (20–119th e.p.p.). Each point represents the results from a single junction. The line from 5 to 150 Hz is drawn through the mean value of the depression at 10, 20, 50, and 150 Hz.

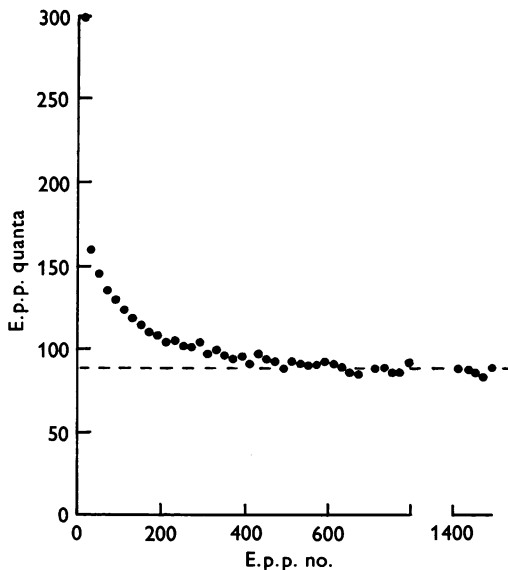


Fig. 5. The quantum output from a neuromuscular junction in a cut muscle preparation during prolonged nerve stimulation. Phrenic nerve stimulation at 150 Hz was continued for a total of 1500 stimuli. Abscissa: the sequence of e.p.p.s in the train of 1500 e.p.p.s. Ordinate: quantum content of the e.p.p.s. Each point represents the mean output from 20 consecutive e.p.p.s, except for the 1st e.p.p.

to those indicated in Figs. 4 and 5, that is, maintenance of quantum content during stimulation at rates below 10–20 Hz, and fall in quantum content during the first 500 stimuli at higher frequencies. During the prolonged stimulation periods no systematic changes in quantum size were observed.

TABLE 4. Quantum release from single junctions in cut muscle tested with two to five different frequencies of stimulation for 1500 impulses. Quantal release values are given for the responses to the first stimulus and the average response to the indicated stimulus numbers

Fibre	Trial	Frequency (Hz)	E.p.p. 1	T_1		T_2	T_3
				20–39	40–119	700–799	1400–1499
B	1	50	201	92	99	112	102
	2	100	216	84	87	68	64
	3	5	225	95	98	101	107
C	1	2	316	225	223	265	227
	2	10	313	197	196	184	177
E	1	50	392	222	189	97	79
	2	100	313	184	147	64	55
H	1	10	280	185	186	182	188
	2	20	255	153	157	152	147
J	1	150	370	256	249	157	141
	2	20	472	358	372	340	297
	3	5	379	309	326	347	359
	4	50	410	327	328	288	237
	5	2	395	404	427	410	387

Effects of tubocurarine

Our results (Tables 3 and 4 and Figs. 1, 4 and 5) obviously suggest that the presence of tubocurarine impairs the ability of neuromuscular junctions to release transmitter, particularly during repetitive stimuli. We therefore examined the effect of a range of concentrations of tubocurarine upon e.p.p. parameters. In preliminary experiments we found that, as Beranek & Vyskocil (1967) had previously reported, in concentrations of $1\text{--}4 \times 10^{-8}$ g/ml. tubocurarine does not change quantum content. A fall in q was, however, detectable which could be as large as 25 %. Our results thus did not support those investigators who suggest a presynaptic site as the primary site of action of tubocurarine on transmitter release.

In the presence of tubocurarine 8×10^{-8} and 4×10^{-7} g/ml., however, presynaptic effects were noticed (Fig. 1C). As Table 3 indicates, with the lower concentration there was a fall in quantum content of the first and the tail e.p.p.s. It is noteworthy, however, that the presynaptic effects were small ($P > 0.05$ in t test) with tubocurarine at 8×10^{-8} g/ml. With

the higher concentration the effects on quantal release measurements (m , t , n), the probability of release (p) and the level of depression (t/m) were very significant ($P < 0.01$ in t test). These parameters were similar to those observed in uncut curarized preparations (Hubbard & Wilson, 1970; Hubbard *et al.* 1971) and in curarized human muscle (Elmqvist & Quastel, 1965a).

It should be noted that we did not find that tubocurarine caused any intermittent failure of release as suggested by Galindo (1971).

Time course of tubocurarine action

In three experiments the time course of tubocurarine action was investigated by recording continuously from one end-plate before, during and after tubocurarine administration (4×10^{-7} g/ml.). One such experiment was reported and illustrated in our preliminary communication (Hubbard *et al.* 1969). In all three experiments of this type, it was found that the parameter reflecting a post-synaptic effect, q , decreased within the turnover time of the bath (4 min). Significantly, however, m and t , which began to decline in the same period, continued to decline for a further 10–15 min. Changes in q and t and m were fully reversible but in confirmation of a difference in the time course of the processes mediating q on the one hand and t and m on the other hand, the recovery times of these parameters were again different, that of q being 5–7 min while that of t and m was some 4 times longer. In these experiments n did not fully recover control amplitude in spite of an hour's washing, perhaps indicating that the change in this parameter reflects an intracellular action of tubocurarine.

Several actions of tubocurarine have been reported (see e.g. Riker & Okamoto, 1969) which could account for the presynaptic actions. The most plausible include a reduction in the presynaptic membrane potential (Hubbard, Schmidt & Yokota, 1965), competitive inhibition of choline uptake (Martin, 1969) and a reduction of ACh synthesis (Bhatnager & MacIntosh, 1967). We tested one possibility – inhibition of choline uptake by examining e.p.p. parameters in uncut diaphragms in the presence and absence of an amount of choline (1.0×10^{-5} g/ml.) which should have been sufficient to overcome any competitive inhibition. This concentration of choline increased the mean quantum content of m from 150 ± 11 (mean $\pm$ S.E., $n = 28$) to 161 ± 16 quanta ($n = 13$) and t from 53 ± 3 to 66 ± 6 quanta but the effect was not significant ($P > 0.05$ in t test). Similarly the changes in n from 1401 ± 149 to 1438 ± 179 quanta and p from 0.107 ± 0.013 to 0.112 ± 0.009 in the presence of choline were not significant. The changes did not mimic the state seen in cut muscle in the absence of tubocurarine (Table 3).

DISCUSSION

In agreement with previous investigators (Lilleheil & Naess, 1961; Lilleheil, 1965) we have found (Figs. 1 and 2) e.p.p. amplitudes better maintained during repetitive stimulation in the absence of tubocurarine. These differences are of the type expected if tubocurarine decreased m , n , and t . Our results (Table 3) indicate tubocurarine does indeed have these effects, and that the pre- and post-synaptic effects of tubocurarine may have different time courses. It is noteworthy that the post-synaptic effect of tubocurarine appears at a lower concentration than does the pre-synaptic.

The question arises, are our results invalidated by the methodological error suggested by Auerbach & Betz (1971)? First, it must be said that this error would alter the absolute value of our quantum content values but would not alter the form of the tetanic response. Secondly, our precaution of recording at sites where the rise time of e.p.p.s was 1.1 msec or less appears to insure that the error was not important for we have shown (a) that m.e.p.p. amplitudes and analysis of variance give very similar values of q (Fig. 3), (b) that changes in e.p.p. amplitude produced by changing [Ca] did not significantly alter any of our experimental parameters (Table 2) and (c) that recording at sites up to 150 μ m from the end-plate focus does not distort quantum release estimates. Auerbach & Betz (1971) appear to have done preliminary work on a technique which could be used to settle the question. They report that when they measured m in their voltage clamped preparation by comparing end-plate currents with miniature end-plate currents they obtained results indicating that the clamp was inadequate. The clamp became adequate in the presence of an unspecified concentration of tubocurarine. We suspect that if the tubocurarine concentration had been further increased in this simplified situation, Auerbach & Betz might have noticed a reduction of m .

The need for a precise estimate of the equilibrium potential is not considered to be a serious problem in this study since the true equilibrium potential appears to vary from fibre to fibre in all preparations (Takeuchi, 1963) and an overestimate of the equilibrium potential by as much as 3 mV has little effect on the quantal release estimates. Furthermore, it is unlikely that the equilibrium potential for the cut muscle preparation is different in the presence and absence of tubocurarine. During prolonged stimulation an alteration in the equilibrium potential due to a change in intracellular [Na] and [K] can be ruled out on the basis that no systematic changes in quantum size (a post-synaptic event) were observed.

Examination of the literature indicates that previously reported investigations of the effect of tubocurarine on ACh release from mammalian

preparations are fully compatible with our results. Investigators have looked at quantum content of e.p.p.s in mammalian preparations in the presence of a range of tubocurarine concentrations which are either sub-threshold for presynaptic effects (4×10^{-8} g/ml., Beranek & Vyskocil, 1967) or well above the level at which presynaptic effects appear fully developed ($4-8 \times 10^{-7}$ g/ml., Bauer, 1971). Only with cut muscle preparations is it possible to examine tubocurarine concentrations between these levels. Again, while most bio-assays for ACh released by nerve stimulation from mammalian nerve-muscle preparations have not shown any reduction of release in the presence of tubocurarine it is clear that the assays are normally not sensitive enough to reveal the 30–50 % fall in release we report (Table 3). Mitchell & Silver (1961) for instance found that a 500–700-fold increase in spontaneous ACh release increased the ACh detected by them only 3 times. Beani *et al.* (1964) conducted the only investigation in which the effects of tubocurarine were assessed using somewhat similar stimulation rates (50 Hz) and temperature parameters (33° C) to those of the present investigation. Significantly, they did find a marked reduction of ACh release in the presence of 5×10^{-6} g/ml. Finally, Beranek & Vyskocil (1967) reported that ACh potentials of denervated diaphragm muscle were less sensitive to tubocurarine than e.p.p.s in innervated diaphragms which could be explained by a presynaptic action of tubocurarine.

At the cat neuromuscular junction the effects of tubocurarine on the form of e.p.p.s and on their quantum content are similar to those we report here (Blaber, 1970; Jacobs & Blaber, 1971). In the frog it has been reported that tubocurarine reduces ACh release from the motor nerve terminal (Gergis, Dretchen, Sokoll & Long, 1971). Auerbach & Betz (1971), however, conclude that low concentrations of TC have no presynaptic effects on the frog. Since the presynaptic actions of tubocurarine are not evident in the rat in low concentration, a higher concentration of tubocurarine may also be required in the frog to demonstrate an effect on the nerve terminal. Auerbach & Betz (1971) did observe a change in CV of end-plate currents in the presence of a higher concentration of tubocurarine ($3-6 \times 10^{-6}$ g/ml.) although they did not interpret this as a reduction in *m*. The effect of tubocurarine in concentrations normally used on the frog ($5-10 \times 10^{-6}$ g/ml.) need to be carefully examined, however, before it can be conclusively stated that tubocurarine has no effect on the frog motor nerve terminal.

The further question arises as to the heuristic value of our results. We suggest two implications. First, we provide an explanation for the conflict, between workers with *in vivo* preparations and workers with *in vitro* curarized preparations, about ACh release during prolonged nerve stimulation. For instance, using curarized guinea-pig and cat and rat prepara-

tions, a marked depression of e.p.p. amplitudes has generally been noted after a few hundred stimuli at rates between 2 and 20 Hz (Brooks & Thies, 1962; Jones & Kwanbunbumpen, 1970; Robbins & Fischbach, 1971). Elmqvist & Quastel (1965*a*) for example, report e.p.p.s reduced to a single quantum after 95 sec of stimulation at 20 Hz. In contrast, numerous reports indicate that *in vivo* muscle preparations are resistant to fatigue at moderate rates of stimulation (Adrian & Bronk, 1928, 1929; Smith, 1934; Bigland & Lippold, 1954; Emmelin & MacIntosh, 1956). Motor units in the human biceps muscle for instance are capable of maintaining activity at the rate of 7–10 Hz for over 14 min (Smith, 1934). Krnjević & Miledi (1958) also found that *in situ* preparations are more resistant to fatigue than *in vitro*.

A similar contrast is apparent in bio-assay studies. In curarized preparations ACh release has been reported to be exceedingly low at rates above 2–5 Hz (Krnjević & Mitchell, 1961; Fletcher & Forrester, 1970). Yet Potter (1970) who assayed for ACh release from the rat diaphragm using a non-curarized preparation found no significant difference in transmitter output when the frequency of stimulation was varied between 1–20 Hz for a brief period of stimulation (total of 360 stimuli). The finding that in the cut muscle preparation e.p.p. amplitude and quantum contents are well maintained (Table 4 and Fig. 4) during nerve stimulation at rates below 10–20 Hz suggests that it is the presence of tubocurarine which prevents *in vitro* curarized preparations from mimicking *in vivo* preparations.

Secondly, our results (Table 4) enable realistic estimates of the number of molecules of ACh in a quantum and hence, according to the vesicle hypothesis, the number to be packed into a vesicle. The maximum number of ACh molecules that can be accommodated inside a vesicle with a 400 Å inside diameter, is 45,000 (Canepa, 1964; Hubbard, 1970). Estimates of the amount of ACh required to produce a one quantum response equivalent to a m.e.p.p. range, however, from 57,000 to 450,000 molecules (Straughan, 1960; Krnjević & Mitchell, 1961; Miledi, 1961; Elmqvist & Quastel, 1965*b*; Potter, 1970). With the exception of the results of Elmqvist & Quastel (1965*b*), these high values were derived by determining the ACh released by 360–1500 stimuli at 1–20 Hz by an assay method, and then dividing this value by the mean quantal release rate, determined from a curarized preparation, estimated by Krnjević & Mitchell (1961) at about only 20 quanta per impulse. As Table 4 indicates we found that the release rate/impulse was much higher in the absence of tubocurarine. Specifically, after an initial frequency dependent decline in amplitude and quantum content (Fig. 2) we found for the e.p.p. generated by stimuli numbers 100 to 1500 at 5 Hz a mean quantal release rate of 199 ± 67 (mean ± 1 s.e., $n = 3$) which rose to 334 ± 59 (mean ± 1 s.e., $n = 3$) for the similar

responses to stimulation at 2 Hz. Using these figures with Potter's (1970) estimate of 4.2 ± 10^6 molecules of ACh per impulse per terminal for rat diaphragms in the absence of tubocurarine at 37° C, we calculate between 12,000 and 21,000 molecules of ACh per quantum, which can easily be packed into a vesicle.

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